Value proposition for statisticians and other quantitative scientists in AI-enhanced drug development

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EFSPI statistical methodology leaders.

Thought experiment: Your CMO automates entire development and reporting pipeline using AI.

Does she miss something?

If no - find another job.

If yes - that is your value proposition.

Let's work out that value proposition.

Indeed, many statisticians find it hard to articulate their own value.

While aspects of the statistician's role will change, I propose that AI will make the core components of a statistician's skillset even more critical in the future.

Harbron (2026)

We have been there before

Modelling & simulation.

Real-world data.

Data Science.

Statisticians are still here.

But this time **might** be different.

> *The frustration of statisticians not always being included in the right discussions isn't a new problem, but without careful management, it could be heightened by the ubiquitous availability of AI.*
>
> *Harbron (2026)*

Value of primarily operational work will change

Activities that **never were a good value proposition for statisticians**:

- Only being able to design a 2-arm RCT with a logrank test.
- Relying on (clinical) trial data alone for drug development.
- Writing lots of ill-documented bandworm code that only runs on your laptop.
- Just executing what the project team demands.
- Playing gate-keeper for analyses.
- Manual output generation, QC.

You may like the **process**: programming, data analysis, pure execution, etc.

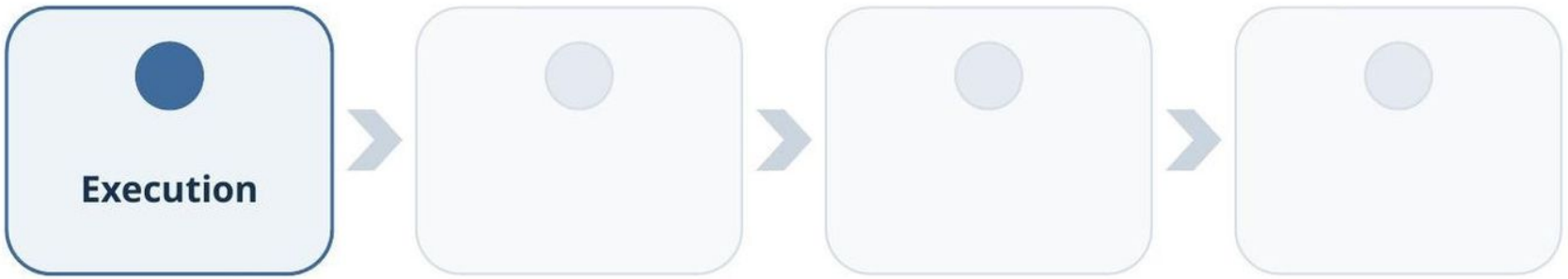
What counts is the **outcome**: impact drug development, **human judgment**.

AI can efficiently generate an answer.

Judgment determines whether it is an answer we should act upon.

The value shift

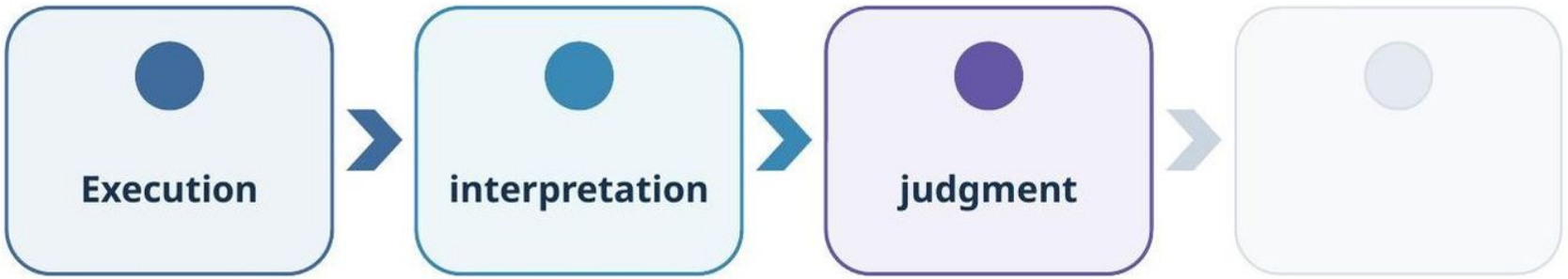
The value shift



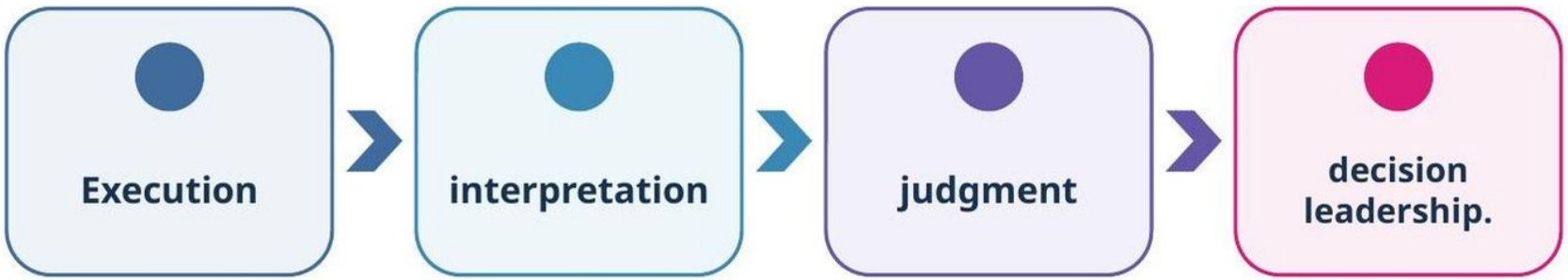
The value shift



The value shift



The value shift



AI drives cost of execution to zero.

Value migrates to the right.

AI increasingly can:

- Generate code.
- Run simulations.
- Draft protocols and SAPs.
- Generate TLFs.
- Identify patterns.
- Summarize results.
- Generate designs.

Humans (statisticians) must still own:

- **Question**. Estimand. Assumptions.
- Trade-offs: Does an answer **make sense**?
- Interpretation and context.
- **Balancing** competing risks and objectives.
- Regulatory argument and **accountability**.
- Taking **responsibility** for the decision.

We also need to address the paradox:

AI may destroy the mechanism by which we acquire judgment.

AI makes experts more productive while potentially making it harder to create next generation of experts.

Lots of discussion

Harbron (2026).

Vandemeulebroecke *et al.* (2026).

Gorst-Rasmussen (2026).

Sosa and Betancourt (2026).

Donoho *et al.* (2026).

Bühlmann (2026).

Hoerl (2026).

Bhattacharya *et al.* (2026).

Where does that leave us?

- **AUTOMATE EXECUTION**: Anything that can safely be automated should be.
- **OWN THE QUESTION**: Do not allow AI efficiency to substitute for problem formulation.
- **INCREASE EFFICIENCY AND OWN INNOVATION**: Use AI to improve trial design and evidence generation.
- **OWN ACCOUNTABILITY**: A machine cannot be accountable to a patient, governance committee, or regulator.
- **COMMIT TO TRAINING THE NEXT GENERATION OF *JUDGES***: Do not allow automation to eliminate the experiences from which expertise develops.

To keep their value statisticians have to move from *backyard* to *frontyard*.

Freely adapted from Tukey and Bühlmann (2026).

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Thank you for your attention.

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Backup

Will there be less work for statisticians?

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Will there be less work for statisticians?



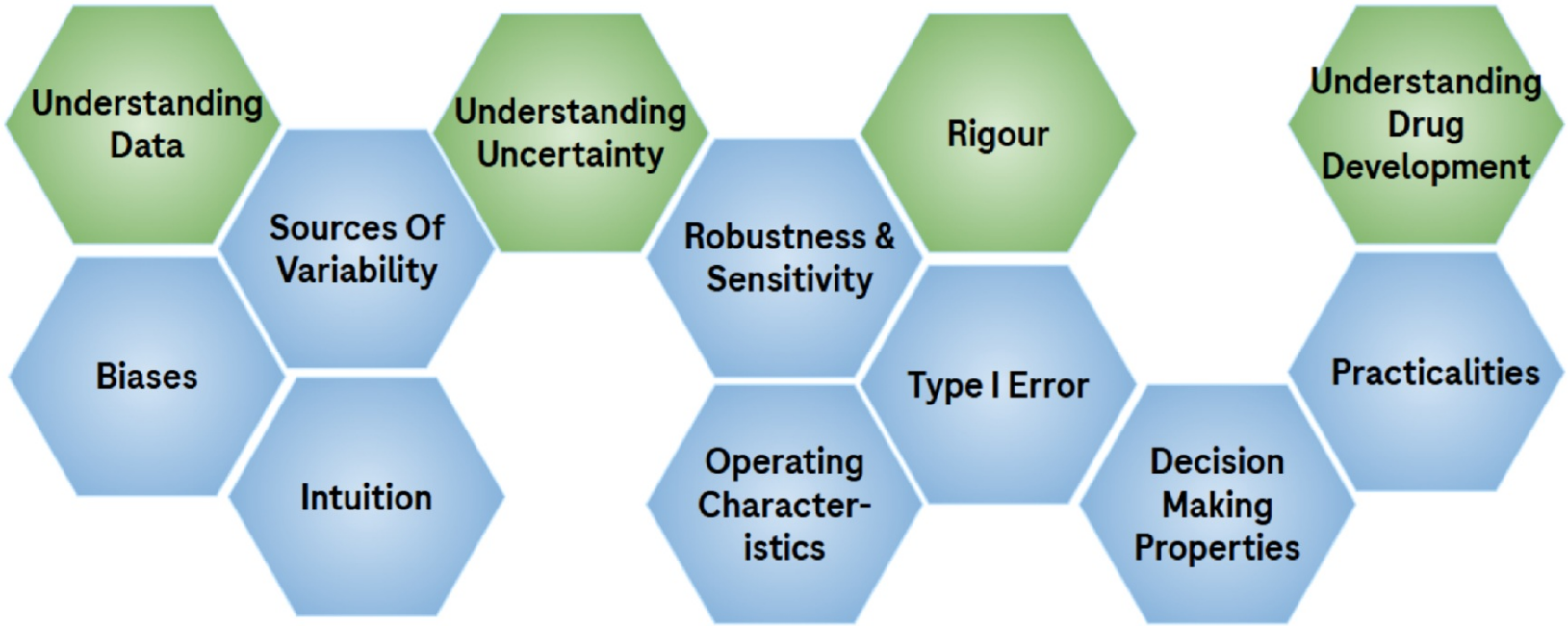
Will there be less work for statisticians?



At least we need better predictions.



What do statisticians bring to the table?



...displaying the **integrity** *to stand by the correct answer even if it is unpopular and not what the team wanted to hear.*

Harbron (2026)

What do statisticians bring to the table?

Statistical judgment is not a checklist, but rather a pattern recognition capacity built over years of encountering failures, anomalies and edge cases. In AI-assisted environments, **statistical judgment functions as the human-in-the-loop** *mechanism that evaluates whether automated outputs align with clinical intent, causal logic and regulatory expectations.*

...

...biostatisticians retain responsibility for **methodological choices, interpretation and regulatory accountability**. *Foundational decisions around study design, estimand definition and overall analysis strategy require expert judgment to* **balance statistical rigor, clinical relevance, ethical considerations and regulatory expectations**.

Bhattacharya *et al.* (2026).

AI is a tool

Does it solve problems it **can solve**, or problems **we have**?

Statisticians can guide that. And evaluate outcome.

Bottlenecks may simply shift: we were never short of drug candidates. Clinical trials are the bottleneck.

> *Use AI to raise the bar for productivity, efficiency, problem-solving, reliability.*
>
> *Frank Harrell*

Jevons' paradox

Technological improvements that increase the efficiency of a resource's use lead to a **rise**, rather than a fall, in total consumption of that resource.

...there will be some occupations for which the three conditions for the paradox will be met, thereby causing increased employment in those fields, such as **radiologists**, **translators**, *and* **coders**.

Erik Brynjolfsson (Stanford prof)

What do AI consultancies say about statistics?

While AI enables gains in speed and scale, it also risks eroding the statistical judgment that underpins scientific credibility, and regulatory confidence.

...

AI does not diminish the need for biostatistical expertise. As routine tasks are automated, the role shifts toward interrogating assumptions, adjudicating ambiguity and translating evidence into clinically meaningful and regulatory-defensible conclusions. These functions cannot be delegated without compromising evidentiary integrity.

...

This requires intentional investment in building statistical judgment, engaging regulators to enable innovation, repositioning methods groups as scientific co-designers, and expanding skill sets to include AI literacy, regulatory science and communication. Organizations that balance digital acceleration with human capability development will define the future standard for rigorous, trustworthy evidence generation.

Bhattacharya et al. *(2026)*.

Towards a value proposition for statisticians and other quantitative scientists in AI-enhanced drug development

Marc Vandemeulebroecke, Mouna Akacha, Klaus Kähler Holst, Michael Krams, Baldur Magnusson, Aris Perperoglou, Kaspar Rufibach, David Wright, on behalf of the EFSPI Statistical Methodology Leaders

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Six tenets

1. Use AI, automate what can be automated.
2. Double down on tasks that need **human judgment**.
3. Collaboration, communication and consultancy skills; ask the right and difficult questions.
4. Drive innovation in statistical and other quantitative methodologies beyond AI.
5. Statistician are used to tension between *curiously engaging with new methodologies* and *firmly holding line of rigor and integrity of robust inference*.
6. Develop future quantitative scientists.

Vandemeulebroecke *et al.* (2026).

Thinking has further evolved.

Analysts not engaged in lifelong learning are already replaceable by AI.
Frank Harrell

R version and packages used to generate these slides:

R version: R version 4.4.3 (2025-02-28 ucrt)

Base packages: stats / graphics / grDevices / utils / datasets / methods / base

Other packages:

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